

Take-Home Assignment: SVM, KNN & Full Week Comparison

Day 33 | PM Session | Week 6, Machine Learning & AI

Topics Covered	SVM kernel selection, C/gamma tuning, KNN optimal K, algorithm comparison methodology, all Week 6 algorithms
Estimated Time	60-90 minutes
Submission	GitHub commit + Jupyter notebook pushed to repo, link in Slack #daily-standup
Due	Next day 09:15 AM

Part A: Concept Application (40%)

Scenario: Build a comprehensive ML cheat sheet notebook. For each of the 8 algorithms learned this week, create a 'card' with: when to use, key hyperparameters, pros, cons, and a 5-line code snippet. Then test all 8 on a dataset of your choice.

Tasks

1. Create an 8-algorithm comparison notebook
2. Each algorithm gets: 1 paragraph description, key params, 5-line code example
3. Run all 8 on one dataset with proper CV
4. Rank and recommend the best model with reasoning

Expected Deliverable

```
Algorithm Card Example:
=== XGBoost ===
When: Tabular data, need best accuracy
Params: n_estimators, learning_rate, max_depth
Pros: Fast, handles missing values, regularized
Cons: Less interpretable than LR/DT
Code: xgb.XGBClassifier(n_estimators=200).fit(X,y)
```

Part B: Stretch Problem (30%)

Self-Study: SVM for Text Classification (TF-IDF + SVM Pipeline)

5. Load a text dataset (20newsgroups from sklearn)
6. Build a TF-IDF + LinearSVC pipeline
7. Train and evaluate on 4 news categories
8. Compare with Logistic Regression on the same text data

Why This Matters

SVM with linear kernel is the CLASSIC text classification algorithm. Used at Google, Yahoo, and Bloomberg for news categorization.

Part C: Interview Ready (20%)**Q1: Conceptual**

Question: You have a dataset with 100 features and 50 samples. Which algorithms from this week would work best and why? Which would fail?

Q2: Coding

Question: Write a function `model_selection_report(X, y, models_dict)` that: splits data, scales, runs CV for each model, returns a formatted DataFrame, and identifies the statistical best model (using paired t-test on CV fold scores).

Q3: Analyze

Question: Your SVM(RBF) has train accuracy 1.0 and test accuracy 0.52. What is wrong? Propose 3 specific fixes.

Part D: AI-Augmented Task (10%)

Task: Use AI to generate a 'decision tree for algorithm selection' — given dataset characteristics (size, features, linearity, labels), which algorithm to use.

Steps

9. Prompt with specific algorithm families covered this week
10. Verify each recommendation against your experience
11. Identify edge cases the AI missed
12. Save as your personal algorithm selection guide

Evaluation Rubric

Criteria	Weight	Excellent	Needs Work
Correctness	40%	All 8 algorithms correctly implemented. Comparison is fair (same CV, same scaling).	Missing algorithms. Unfair comparison.
Code Quality	25%	Clean, modular notebook. Reusable for any dataset.	Messy. Not reusable.
Understanding	20%	100-feature analysis shows depth. SVM overfitting diagnosis is thorough.	Shallow analysis.

AI Usage	15%	Algorithm selection guide is verified and improved.	Unverified.
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